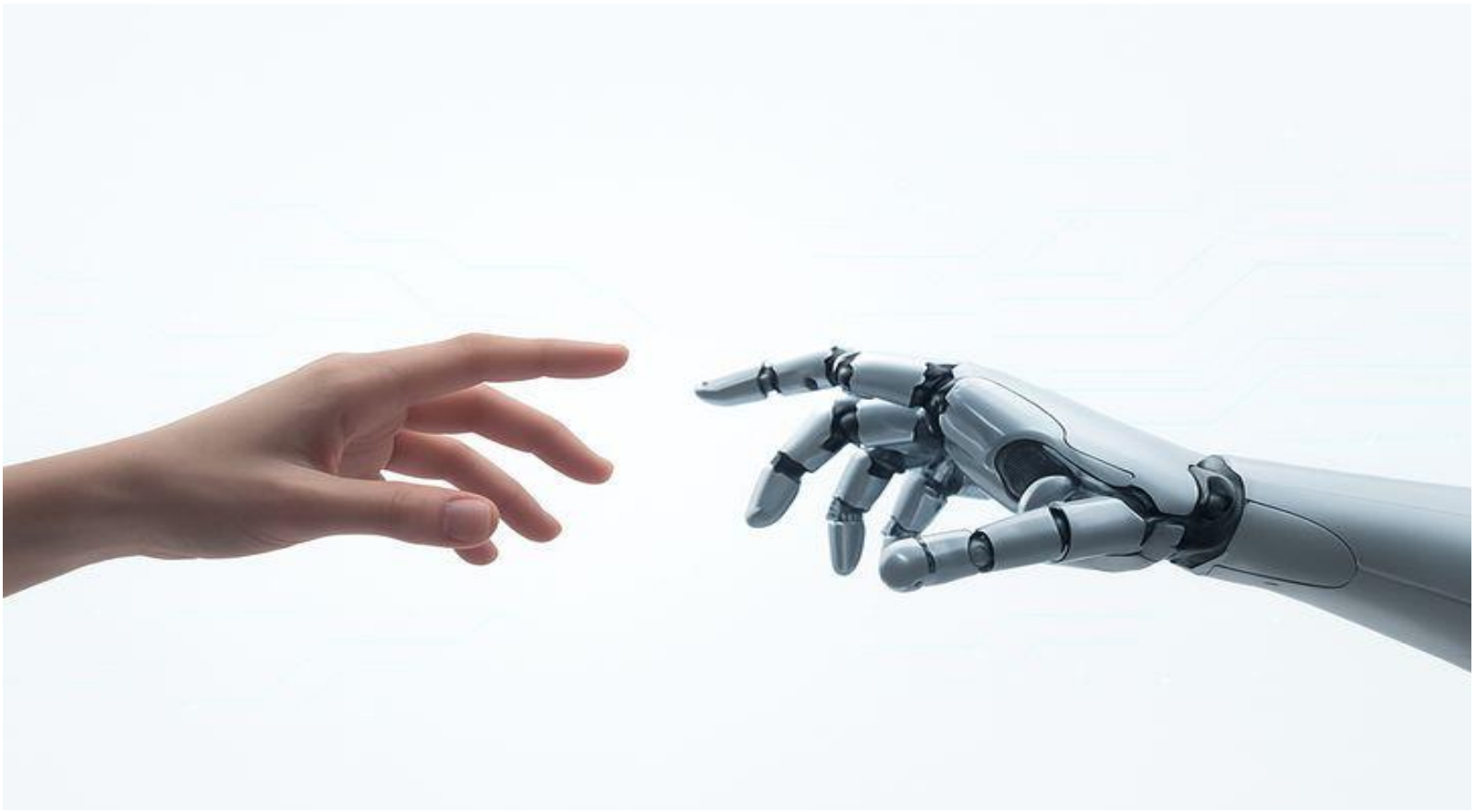


Head & Hand

AI Career & Interview Coach

HCAI Term Project — Milestone 1

HCAI Principle: User Control and Agency

TEAM MEMBER:

- Nora Walid Ibrahim
- Maisoun Ebrahim Ahmed
- Noor Adal Mohammed
- Fatma Saber Elgendy

Prompt

I'm working on a project for a Human, Computer, and AI Interaction (HCAI) course. The project involves building and evaluating a web-based AI chatbot for career and interview coaching.

Our chosen HAI principle is **User Control and Agency**, which means users can accept, reject, modify, or undo the AI's suggestions and actions.

The AI acts as an active coach that guides users, asks questions, provides feedback, and supports them in making their own decisions rather than simply answering questions.

OUR CHATBOT SHOULD SUPPORT THESE SCENARIOS

1. CV creation/improvement: The user says something like "I want to make a CV, what do I do?" or "improve my CV". The **coach** asks step-by-step questions (education, experience, skills, projects, target job) to build the CV with the user.
2. Mock interview: The user asks for an interview. If the user doesn't specify the job track/field, the coach asks for it (plus level, experience, company type), then runs a realistic interview one question at a time.
3. Post-interview feedback: After the interview, the coach gives FAIR, rubric-based feedback: strengths, weaknesses, what to improve, and what to study/practice to fit the target job.
4. Adding to an existing CV: If the user has a CV and wants to add something (a course, project, skill) but doesn't know where it goes, the coach tells them which section it belongs in, explains why, and writes it.
5. Job-fit check: The coach compares the CV against a target job description and gives a match percentage with a clear breakdown of how it was calculated.
6. CV evaluation: The coach gives a CV score based on a visible rubric (e.g., structure, clarity, relevance, impact, language, formatting).

CONSTRAINTS

- Use only public/synthetic/fictional data. No real personal data.
- Must include fairness: feedback must not change based on name, gender, nationality, or language. We will test this with swapped names/genders.
- Must handle failures: out-of-scope questions, weak info, API errors, with clear fallback messages.
- Must be accessible (keyboard use, screen reader friendly, plain-language option).
- Users may write in Arabic or English.
- Explanations must reflect what the system actually did (no invented reasoning).

YOUR TASK

Draft ONE concept for the core interaction flow of this chatbot. Include:

1. Concept name and a one-paragraph summary.
2. Target user group (be specific, to make our idea unique).
3. The conversation flow: a step-by-step flow for each of the 6 scenarios above (what the coach says/asks, what the user can do).
4. User Control and Agency features: at least 8 concrete features (e.g., edit answers, accept/reject each suggestion, undo, choose feedback style, skip a question, choose interview difficulty, approve before CV is changed, adjust the rubric weights). For each one: where it appears in the flow and how it works.
5. How the coach scores things: the CV rubric, the job-fit % formula, and the interview feedback rubric. Make them transparent and fair.
6. Error recovery, fallback, and human handover moments.
7. A simple text flow diagram (using arrows) of the whole conversation.
8. A "HAI Principle Ladder" with 4 levels: Inform, Explain, Allow Control, Enable Feedback. Give one feature and one measurable target per level.
9. Strengths and weaknesses of your own concept (be honest).
10. A short sample dialogue (8-10 turns) showing the coach in action.

Functional Workflow Architecture of CareerCompass AI (ChatGPT)

1. Initial Engagement & Scenario Identification

- System Entry: The user opens CareerCompass AI.
- Role Clarification: The AI explains its role and supported tasks.
- Task Selection: The user selects a career task.
- Scenario Identification: The system identifies the appropriate scenario for the selected task.

2. Workflows & Scenario Execution

Based on the user's choice, the workflow branches into 6 primary scenarios:

1. CV Creation / Improvement

- Ask about target role, education, skills, and projects.
- Check information and identify missing details.
- Evaluate if there is enough reliable information:
  - If No: Ask a focused question or allow skipping.
  - If Yes: Proceed to generate a draft or recommendations.

2. Mock Interview

- Ask about role, level, and interview difficulty.
- Ask one interview question at a time.
- User answers.
- Provide feedback and the next question.
- Check whether to continue the interview:
  - If Yes: Loop back to ask another question.
  - If No: Evaluate answers using a visible rubric, explain scores with evidence, and present feedback and an improvement plan.

3. Post Interview Feedback

- Collect interview answers or transcript.
- Evaluate answers using a visible rubric.
- Explain scores with evidence.
- Present feedback and an improvement plan.

4. Update Existing CV

- Ask which CV section or detail to update.

- Check information and identify missing details, then ensure sufficient information is available to generate a draft.

5. Job Fit Analysis

- Collect CV and job description.
- Compare job requirements with CV evidence.
- Show match score, evidence, and unverified gaps.
- Present job-fit analysis.

6. CV Evaluation

- Collect CV and identify target role.
- Evaluate CV using a visible rubric.
- Show CV score, rubric breakdown, and suggestions.
- Present CV evaluation.

3. Draft Generation & Presentation

- Generation: Generate a draft or recommendations based on gathered data.
- Presentation: Present draft for user review.

4. User Decision & Revision Handling

The user makes a decision based on the presented draft:

- Reject: Keep original content unchanged.
- Undo: Restore previous state where supported.
- Edit / Correct: Revise based on user feedback, then present the draft again for user review.
- Accept: Move forward to confirmation and application.

5. Final Confirmation & Execution

- Confirmation: Confirm before applying any changes.
- Execution Status Check: Check if the application confirms success:
  - If Yes: Show updated result.
  - If No: Show error and offer retry or manual recovery.
- Completion: End the session or choose another task

Functional Workflow Architecture of CareerCompass AI(Gemini)

1. Initial Engagement & Goal Clarification

- System Entry: The user opens CareerCompass AI and is greeted with a welcome message and a list of supported tasks.
- Task Selection: The user selects a specific career task.
- Goal Verification: The system checks if the goal is clear and supported:
  - Unclear / Vague Goal: The system asks a clarifying question.
  - Unsupported Goal: The system explains limitations and redirects the user.
  - Clear & Supported Goal: The system advances directly to selecting the appropriate workflow.

2. Workflows & Data Collection

Based on the selected path, the system branches into 6 primary workflows:

- Create or Improve CV: Collects required information topic by topic.
- Mock Technical Interview:
  - Confirms role, level, experience, and difficulty.
  - Asks one interview question at a time.
  - Evaluates answers and prepares feedback.
- Post-Interview Feedback:

    - Collects interview answers or transcripts.
    - Evaluates answers using a visible rubric.
- Update Existing CV: Identifies the target section and requested changes.
- Analyze Job Fit: Collects the candidate's CV and the job description.
- Evaluate CV: Collects the CV and the target role.

3. Draft Generation & Output Display

- Processing: The system generates a draft or analysis based on the collected inputs.
- Result Presentation: Displays results accompanied by supporting evidence and explanations.
- Review Interface: Displays a review panel alongside the main conversation window.

4. Review & Decision Handling

The user reviews suggestions provided by the system:

- Inline Edit:
    - Edits suggestions directly inline.
    - Regenerates or revises the affected parts.
- Reject: Discards selected suggestions.
- Accept: Marks selected suggestions as accepted.

5. Final Confirmation & Execution

- Readiness Check: Verifies whether changes are ready to be applied.
- Preview & Confirmation:
    - Displays a final preview and requests confirmation.
    - Option to undo/redo or restore/reapply a previous state if needed.
- Execution & Status:
    - Applies approved changes upon confirmation.
    - Success Path: Displays the final confirmed result.
    - Failure Path: Shows an error message, preserves existing work, and offers a retry or manual alternative.

Comparison

Comparison Criteria	ChatGPT	Gemini
Clarity of Workflow	Strong; clearly defines steps, branches, and decisions.	Good; focuses on the main steps.
Handling Insufficient Information	Specifies what happens when information is insufficient.	Requires additional details on how insufficient information is handled.
Control Over AI Suggestions	Allows users to accept, reject, edit, and undo changes.	Allows users to selectively accept or reject suggestions and edit them directly.
Suggestion Review Interface	Provides less detail about the review interface design.	Stronger; includes a dedicated review panel and inline editing.

Error Handling	Defines success, failure, and recovery paths.	Emphasizes preserving previous work and offering retry options.
Ease of Implementation and Testing	Easier to implement and test due to its clearly defined workflow logic.	Offers a clearer review experience but requires further specification of some workflow rules.
Suitability for Our Project	The preferred foundation for the project.	A good alternative, but some details need further development.

Weaknesses& Strengths

Comparison	ChatGPT Draft	Gemini Draft
Strengths	Clear steps, handles missing information, and explains decisions and errors.	Clear review panel, direct editing, and easy acceptance or rejection of suggestions.
Weaknesses	Less detail about the review interface and editing individual suggestions.	Less detail about missing information and workflow decisions.
Main Focus	How the system works.	How the user interacts with the results.
Main Advantage	Easier to follow and implement.	Easier to review and edit suggestions.

Chosen Direction: ChatGPT Draft

After comparing both drafts, we selected the ChatGPT draft as the foundation for CareerCompass AI.

We chose this direction because it provides a more detailed functional workflow, with explicit decision branches, information-sufficiency checks, and clear handling of user decisions and execution errors. These details make the workflow easier to implement, test, and evaluate within our project constraints.

Although the Gemini draft provides a more explicit review interface and inline editing experience, we believe the ChatGPT draft offers a stronger overall foundation for defining the system's behavior.

The selected workflow supports our HCAI principle, User Control and Agency, by allowing users to review, accept, reject, edit, and undo AI-generated suggestions before applying changes.

Functional Workflow Architecture of CareerCompass AI (After Updata)

1. Initial Engagement & Scenario Identification

- **System Entry:** The user opens CareerCompass AI.
- **Role Clarification:** The AI explains its role, capabilities, and supported career tasks.
- **Task Selection:** The user selects a career task.
- **Scenario Identification:** The system identifies the appropriate scenario based on the selected task.
- **User Control:** The user can change the selected task or return to the main menu at any time.

2. Workflows & Scenario Execution

Based on the user's choice, the workflow branches into seven primary scenarios.

1. CV Creation / Improvement

- Ask about the target role, education, skills, projects, and relevant experience.
- Check the provided information and identify missing details.
- Evaluate whether enough reliable information is available.
    - If No: Ask a focused question or allow the user to skip it.
    - If Yes: Generate a CV draft or improvement recommendations.
- Present the draft for user review.
- Allow the user to accept, reject, edit, or undo suggestions where supported.
- Present the updated content after the user's decision.

2. Mock Interview

- Ask about the target role, experience level, and interview difficulty.
- Ask one interview question at a time.
- Allow the user to submit an answer before presenting the next question.
- Provide feedback based on the user's answer when appropriate.
- Ask whether the user wants to continue the interview.
    - If Yes: Ask another question.
    - If No: End the question session and evaluate the answers using a visible rubric.
- Explain scores with evidence and present strengths, weaknesses, and an improvement plan.
- Ask the user what they would like to do next:
    - Practice again, focusing on weak areas.
    - Prepare for a real interview.
    - Return to CV Creation / Improvement or CV Evaluation.
    - Continue the job application journey.
    - Finish the session.
- Allow the user to choose the next step.

3. Post-Interview Feedback

- Ask the user to provide interview answers, a transcript, or relevant notes from a real interview.
- Check whether enough information is available for a meaningful evaluation.
    - If No: Ask a focused question or explain the limitations of the available information.
    - If Yes: Evaluate the answers using a visible rubric.
- Explain scores with evidence from the provided answers.
- Present strengths, improvement areas, and an improvement plan.
- Ask the user whether they want additional practice, preparation for another interview, or help with the next application step.

4. Update Existing CV

- Ask the user to provide the existing CV and identify the section or detail to update.
- Check the available information and identify missing details.
- Ask focused questions when additional information is needed.

- Generate a proposed revision without changing the original content automatically.
- Present the proposed changes for user review.
- Allow the user to accept, reject, edit, or undo changes where supported.
- Apply confirmed changes only when the system supports the operation.
- Present the updated content for review.

5. Job Fit Analysis

- Collect the user's CV and the target job description.
- Extract the main job requirements and relevant CV evidence.
- Compare the user's documented skills, experience, education, and projects with the job requirements.
- Identify matching qualifications, missing requirements, and unverified information.
- Present a match score only when a defined and explainable scoring method is used.
- Explain the evidence behind the analysis and clarify that the result is an estimate, not a hiring decision.
- Suggest relevant improvements or preparation steps.
- Allow the user to revise the CV, prepare an application, or choose another task.

6. CV Evaluation

- Collect the user's CV and target role.
- Evaluate the CV using a visible rubric with defined criteria.
- Present the overall score, rubric breakdown, supporting evidence, and improvement suggestions.
- Explain missing or unclear information without inventing qualifications or experience.
- Allow the user to review, accept, reject, or edit suggestions.
- Offer relevant next steps, such as CV improvement, job-fit analysis, or application preparation.

7. Job Application & Offer Support

- **Job Selection:** Ask the user to identify the target job and provide its description or relevant details.
- **Application Preparation:** Help the user review the CV, identify relevant skills and experience, and prepare application materials such as a cover letter when needed.
- **Application Status:** Allow the user to record or update the current application status:
    - Preparing to Apply
    - Applied
    - Interview Scheduled
    - Rejected
    - Offer Received
- **Next-Step Guidance:** Provide guidance based on the user's reported status:
    - If Preparing to Apply: Review the application materials and help prepare for submission.
    - If Applied: Suggest appropriate follow-up steps or preparation for a possible interview.
    - If Interview Scheduled: Offer mock interviews and role-specific preparation.
    - If Rejected: Help the user review available feedback, identify improvement areas, and prepare for another opportunity.
    - If Offer Received: Help the user review the offer details, compare them with their priorities, and prepare questions or negotiation points.
- **Offer Decision Support:** Help the user consider relevant factors, such as role responsibilities, compensation, benefits, working conditions, and career goals, when this information is available.
- **User Decision:** The user decides whether to accept or decline the real job offer. The AI provides guidance but does not make the decision on the user's behalf.
- **Continue or Finish:** Allow the user to continue with another career task or end the session.

- **Limitations:** The system relies on information provided by the user. It does not automatically track external applications, contact employers, submit applications, make hiring decisions, or guarantee a job offer.

3. Draft Generation & Presentation

- **Generation:** Generate a draft, analysis, feedback, or recommendations based on the selected scenario and the information collected.
- **Output Type:** Present the appropriate output for the selected task, such as:
    - A CV draft or suggested revisions.
    - Interview questions, feedback, or an improvement plan.
    - A job-fit analysis.
    - A CV evaluation report.
    - Application preparation materials.
    - Job-offer review and negotiation preparation.
- **Evidence and Explanation:** Explain recommendations using available information and clearly identify missing or unverified details.
- **Presentation:** Present the output in a clear format for the user to review.
- **User Review:** Ask the user whether they want to accept the suggestions, reject them, request edits, or choose another step.

4. User Decision & Revision Handling

The user makes a decision based on the presented draft, feedback, or recommendations.

- **Reject:** Keep the original content or previous state unchanged where supported.
- **Undo:** Restore the previous state where supported.
- **Edit / Correct:** Revise the proposed output according to the user's feedback and present it again for review.
- **Accept:** Confirm that the user agrees with the proposed content or recommendation.
- **Post-Decision Routing:** Determine the next step based on the selected scenario and the user's choice:
    - If the user accepts a CV revision, present the updated CV content.
    - If the user accepts interview feedback, offer additional practice or relevant preparation steps.
    - If the user accepts an application suggestion, help prepare the next application step.
    - If the user receives a job offer, provide offer-review and negotiation support without deciding for the user.
    - If the user rejects a suggestion, preserve the original content or previous state where supported.
- **Explicit User Confirmation:** Ask for confirmation before applying supported changes to existing content.
- **Decision Boundaries:** Accepting an AI-generated suggestion does not automatically submit an external application or accept a real job offer.

5. Final Confirmation, Execution & Session Routing

- **Confirmation:** Before applying a change to an existing CV or other supported content, ask the user to confirm the intended change.
- **Execution:** Apply the confirmed change only when the system supports the operation. For tasks that generate advice, feedback, or analysis, present the final output without requiring an external action.
- **Execution Status Check:** If an operation is performed, check whether it succeeded:
    - If Yes: Present the updated result.
    - If No: Explain the error and offer a retry or manual recovery.

- **Next-Step Routing:** After completing the current task, offer relevant next steps based on the user's goal, such as improving the CV, practicing an interview, preparing a job application, reviewing interview outcomes, or reviewing a job offer.
- **Session Completion:** Allow the user to choose another task or end the session at any point.
- **External Actions:** The system does not submit applications, contact employers, or accept job offers automatically. The user remains responsible for these decisions and actions.

6. Continuous Career Coaching Loop

- **Iterative Career Journey:** The career journey is not strictly linear. Users can return to previous stages based on their needs and progress.
- **After Interview Practice:** The user can repeat the mock interview, work on weak areas, or prepare for an upcoming real interview.
- **After Application Outcomes:** The user can review available feedback, improve application materials, and apply for other opportunities.
- **After Receiving an Offer:** The user can review the offer, prepare negotiation questions, and decide whether to accept or decline.
- **Continue or Finish:** The system continues to provide relevant guidance while the user chooses to continue. Reaching a job offer is a possible outcome, not a guaranteed result or a mandatory stopping point.
- **User Agency:** The user remains in control of task selection, proposed changes, workflow transitions, and final career decisions.

7. Safety, Transparency & User Control

- **Transparency:** Explain the reasons and available evidence behind recommendations and evaluations.
- **User Control:** Allow the user to accept, reject, edit, or undo AI-generated suggestions where supported.
- **Accuracy:** Do not invent qualifications, skills, experience, interview answers, or job-offer details.
- **Fairness:** Focus on job-relevant qualifications and avoid using protected or irrelevant personal attributes to evaluate candidates.
- **Privacy:** Ask only for information needed for the selected task and handle personal information according to the system's privacy practices.
- **Human Decision-Making:** Do not make hiring decisions or accept job offers on the user's behalf.
- **Realistic Expectations:** Provide coaching and decision support without promising interview success or employment.

Prompt Log — Appendix A

Entry 1 — ChatGPT: Initial Concept

Prompt: Design a user-controlled career and interview coach for fresh computing graduates.

Output: Six scenarios, conversation flows, scoring rubrics, and user-control mechanisms.

Team decision: Narrow the target audience to fresh computing graduates.

Entry 2 — Gemini: Alternative Concept

Prompt: Propose an alternative design for the same project and HAI principle.

Output: Review panel, accept/reject controls, undo/redo, and adjustable interview difficulty.

Team decision: Keep ChatGPT as the primary direction while considering useful UI features from Gemini.

Entry 3 — ChatGPT: Conversation Flow

Prompt: Clarify the chatbot's coaching process.

Output: A step-by-step interaction flow.

Team decision: Ask one question at a time and allow users to skip.

Entry 4 — ChatGPT: Scoring

Prompt: Define transparent scoring criteria for CVs, job fit, and interviews.

Output: Weighted rubrics and evidence-based feedback.

Team decision: Explain scores and label missing evidence as unverified.

Entry 5 — ChatGPT: Boundaries

Prompt: Define unsupported-request and error-handling behavior.

Output: Rules for clarification, redirection, and recovery.

Team decision: Prevent fabricated information and false success claims.

Idea for Approval

1. Project Title

- CareerCompass AI: A User-Controlled Career and Technical Interview Coach for Fresh Computing Graduates

2. Problem Statement

- Fresh computing graduates often have academic projects and technical skills but limited professional experience.
- They may struggle to:
  - Present their achievements effectively on a CV.
  - Determine whether they are suitable for entry-level jobs.
  - Prepare for technical interviews.
- AI tools may generate unsupported claims or provide evaluations without clear explanations.
- CareerCompass AI aims to solve these problems through structured coaching, transparent feedback, and user control.

3. Target Users

- Fresh graduates from Egyptian universities in:
  - Computer Science.
  - Computer Engineering.
  - Information Technology (IT).
  - Artificial Intelligence (AI).
  - Related computing fields.
- Students and graduates preparing for their first technical job or internship.

4. Proposed Solution

- Develop a web-based chatbot that supports six scenarios:

1. CV Creation and Improvement: Help users create or improve their CVs.
2. Mock Technical Interviews: Simulate technical job interviews.
3. Post-Interview Feedback: Provide feedback on interview performance.
4. CV Updating: Help users update an existing CV.
5. Job-Fit Analysis: Compare a CV with a job description.
6. CV Evaluation: Evaluate CVs using clear and visible criteria.

5. Main HAI Principle: User Control and Agency

- The AI provides recommendations, but the user makes the final decision.
- Users can:
    - Edit or reject AI suggestions.
    - Correct misunderstandings.
    - Review scores and their explanations.
    - Approve changes before they are applied.
- The system should keep users in control of their personal information and career decisions.

6. What Makes the Idea Distinctive?

- Focuses specifically on fresh computing graduates.
- Helps users present academic projects and technical skills honestly.
- Provides evidence-based feedback instead of unsupported claims.
- Combines six connected career-coaching scenarios.
- Uses transparent evaluation criteria.
- Prioritizes user control and decision-making.

7. Testing and Evaluation

- Test the prototype using synthetic or publicly available examples.
- Support both Arabic and English.
- Check whether recommendations differ across demographic variations.
- Evaluate the clarity, reliability, and usefulness of the AI's feedback.

8. Feasibility and Expected Outcome

- A team of four to five students can develop the prototype.
- The prototype will include:
    - A chat interface.
    - Six career-coaching scenarios.
    - Transparent scoring explanations.
    - Editable AI suggestions.
    - User approval controls.
    - Basic error handling.
- Expected Outcome:
    - Help fresh graduates improve their CVs.
    - Improve their technical interview readiness.
    - Provide clear and reliable feedback.
    - Ensure users retain control over their information and decisions.